

1.9 User Interface Design (UI vs UX)

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CS XII

Chapter 1: Computer Systems (HCI)

Lecture 12 of 15

★ Golden

1.9 User Interface Design (UI vs UX)

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LEARN IT

UI design is the process of creating and organizing the **visual elements** of a software application so users can interact with it easily and efficiently.

1.9.1 Components of a user interface

Component	Role
Buttons	Clickable elements that perform actions (Submit, Cancel).
Menus	Lists of options for navigating the system.
Icons	Small symbols representing tools or actions.
Text fields	Boxes where users type information (passwords, search).
Checkboxes / radio buttons	Selecting options in a form. Checkboxes allow several choices; radio buttons allow only one in a group.
Sliders	Adjusting values along a bar (volume, brightness).

1.9.2 User experience (UX)

UX is the overall experience a user has while interacting with a system — how smooth, comfortable and satisfying it feels.

UI is what you see; UX is how it feels.

WORKED EXAMPLE

Source example: food-delivery app. The UI is the nice pictures of burgers and the red buttons. The UX is that the app loads fast, tracking works, and the food arrives without the user getting frustrated.

Local walk-through: JazzCash.

- **UI:** green Send button, JazzCash logo, amount text field, fingerprint icon, menu of Bill / Load / Send.
- **UX:** login in two seconds, no crash in a weak-signal bazaar, money arrives, SMS comes, the user feels safe and finished — or, if it hangs with no tick, the UX is bad even if the buttons look pretty.

School portal: UI = logo, "Login" button, date-sheet menu icon. UX = a student finds the chemistry practical date in under a minute on a phone, without calling the clerk.

Pretty UI + slow, confusing flow = poor UX. Plain UI + fast, clear flow = good UX.

REMEMBER IT

Clothes vs day out. UI is the outfit (colours, buttons, icons). UX is the whole day (Did you arrive on time? Did you get wet? Were you stressed?). Examiners love the line: *UI is what you see; UX is how it feels.*

COMMON MISTAKE

Mistake: Writing "UI and UX are the same" or "UX is just colours".

Fix: UI = visual components (buttons, menus, icons, fields, checkboxes, sliders). UX = overall smoothness, comfort and satisfaction, including speed and whether the task succeeds.

اردو میں سمجھیں

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EXAM TIP

Highest-yield short question in the chapter: "Differentiate between UI and UX with an example." Write: (1) the two definitions, (2) the sentence "UI is what you see; UX is how it feels", (3) the food-delivery example from the source, (4) optionally JazzCash or the school portal. For 5 marks also list four UI components. Do not spend the answer describing Figma — that is the next topic.

CHECK YOURSELF

1. List six UI components from the source.
2. In the food-delivery example, what is UX?
3. Complete: "UI is what you see; UX is ____."

Answers: 1. Buttons, menus, icons, text fields, checkboxes/radio buttons, sliders. 2. The app loads fast, tracking works, food arrives without frustration. 3. how it feels.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. UI is:

- a. The look and interactive controls
- b. The whole research journey only
- c. The database schema
- d. The sprint velocity

MCQ 2. UX is:

- a. The overall experience (useful, usable, satisfying)
- b. Only the colour palette
- c. Only the logo
- d. Only the NAND count

MCQ 3. You can have pretty UI and bad UX when:

- a. The task still takes 20 confusing steps
- b. Buttons are clear and short
- c. Feedback is excellent
- d. Accessibility is strong

MCQ 4. UX research includes:

- a. Interviews, tests, personas
- b. Only picking hex colours
- c. Only SQL
- d. Only OSI

MCQ 5. This is ★: define both, then:

- a. Give a product example of each going wrong
- b. Draw a K-map
- c. Write a sort
- d. List Access objects

MCQ 6. A progress tracker in a 3-step form is mainly:

- a. UX (and a bit of UI)
- b. A foreign key
- c. A histogram
- d. A gate

Extra short questions

1. Define UI vs UX.
2. Who typically does each job.
3. Example: ATM.
4. Why both matter in exams.

Extra long question

1. A food-delivery app is gorgeous but checkout fails twice. Analyse UI vs UX, then list 4 UX fixes that may not need new colours. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. UI: visual/interactive layer. UX: how the whole journey feels and whether the goal is achieved easily.
2. UI designer: screens/components. UX: research, flows, prototypes, testing. They overlap.
3. UI: button layout. UX: whether you felt safe and finished withdrawal quickly.
4. Papers ask the difference plus one example — do not mix the letters.

Long

1. UI may be fine. UX: errors, extra steps, unclear fees. Fixes: guest checkout, saved address, fee shown early, retry without losing the cart, better messages.

Next lecture: 1.9.3 – 1.9.5 Wireframing, Figma, and Prototypes